



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch : III Semester B.E CSE B
Course Code & Title : CS3352 & Foundations of Data Science
Name of the Faculty member : Mrs. P.Sabeena Burvin, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit IV / Aggregation and grouping
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 1,2,4,10
- **Activity Chosen:** Role play
- **Date of Implementation:** 09.10.2025
- **Justification:**

Role-playing and scenario discussions require effective communication, both in understanding the situation and conveying one's thoughts and feelings. Among the key benefits of role play is the enhancement of a student's ability to express themselves, which also improves their listening and interpretative skills. Effective communication is a fundamental aspect of positive behaviour, as it allows individuals to articulate their needs and concerns clearly.

- **Time Allotted for the Activity:** 20 minutes
- **Details of the Implementation:**
- **Roles Assigned:**
 - Student 1: Role of the Breetta J, Applying Multiple Aggregations using (agg()) function with suitable python program.
 - Student 2: Role of the Parvathi devi K, responsible for Applying Grouping function in pandas with suitable python program.

Activity Execution:

Student 1 explained the aggregations function implementation by using the following key points with example python program, The agg() function apply for multiple aggregation functions at the same time.

Key Points:

1. **Axis Parameter:** The axis parameter determines the dimension along which the aggregation is performed. axis=0 operates column-wise, and axis=1 operates row-wise in a 2D array.
2. **Performance:** NumPy's aggregate functions are significantly faster on large arrays compared to Python's built-in functions (like sum()) because they are compiled.
3. **Missing Data:** Most functions have a nan safe counterpart (e.g., np.nansum()) that ignores

NaN (Not a Number) values, preventing the entire operation from returning NaN.

Program execution:

```
df.agg(['sum', 'min', 'max'])
```

Output:

	Maths	English	Science	History
sum	24	20	23	20
min	7	4	7	5
max	9	10	8	9

Student 2 explained the grouping function implementation by using the following key points with suitable python program. She explained, grouping in Pandas means organizing your data into groups based on some columns. Once grouped you can perform actions like finding the total, average, count or even pick the first row from each group. This method follows a split-apply-combine process:

- 1.Split the data into groups
- 2.Apply some calculation like sum, average etc.
3. Combine the results into a new table.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO12	PSO1	PSO3
CO1	2	2	3	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO12	PSO1	PSO3
	2	2	3	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of visualization libraries in python to interpret data for solving problems	Identify the need of aggregation on libraries in python to interpret and explore data	Able to design solutions for any applications using grouping on libraries in python	Able to analyze and apply aggregation libraries in python and provide valid conclusions	Ability to recognize the need for aggregation and grouping concepts	Exhibit knowledge for data science application development

- **Images / Screenshot of the practice:**



Figure 1: Breetta J executes aggregation functions



Figure 2: Parvathi Devi K executes grouping functions

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students reported improved confidence in explaining concepts.
- The hands-on experience encouraged even passive learners to engage in the lesson. Active participation promoted a better understanding of the topic. They felt that through such learning, they can explore more.

- ❖ ***Benefit of the practice:***

- Motivate and engage students.
- Learn skills used in real-world situations.
- Learners felt more ownership of knowledge.

- ❖ ***Challenges faced in implementation:***

- The activity involved only two students, which left the rest of the class as observers. It was challenging to ensure that all students could engage in hands-on learning.
- Time management during student-led sessions required guidance.

References:

- ❖ <https://serc.carleton.edu/introgeo/roleplaying/whatis.html>
- ❖ <https://www.niu.edu/citl/resources/instructional-guide/role-playing.shtml>

